



# Marine Biology

# 海洋生物学



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## • Major Phyla of Marine Invertebrates

1. Sponges (海绵动物)
2. Cnidarians (腔肠动物)
3. Flatworms (扁形虫)
4. Ribbon worms (纽虫)
5. Nematodes (线虫)
6. Annelids (环节动物)
7. Sipunculans (星虫动物)
8. Echiurans (螠虫动物)
9. Arthropods (节肢动物)
10. Molluscs (软体动物)
11. Bryozoans (苔藓虫)
12. Phoronids (帚虫)
13. Brachiopods (腕足类)
14. Arrow worms (箭虫)
15. Echinoderms (棘皮动物)
16. Hemichordates (半索海生动物)
17. Tunicates (被囊动物)
18. Lancelets (文昌鱼)

## 7. The Annelids (Segmented Worms, 环节动物)

- About 20,000 species
- Live in salt water, freshwater or moist terrestrial
- Divided into **repeating sections** called **segments with many internal organs**
- **Bilateral symmetry**
- **Setae (刚毛)** in most (except leeches)
  - these are bristle-like structures that extend from the sides of each segment which help the annelid stay in place

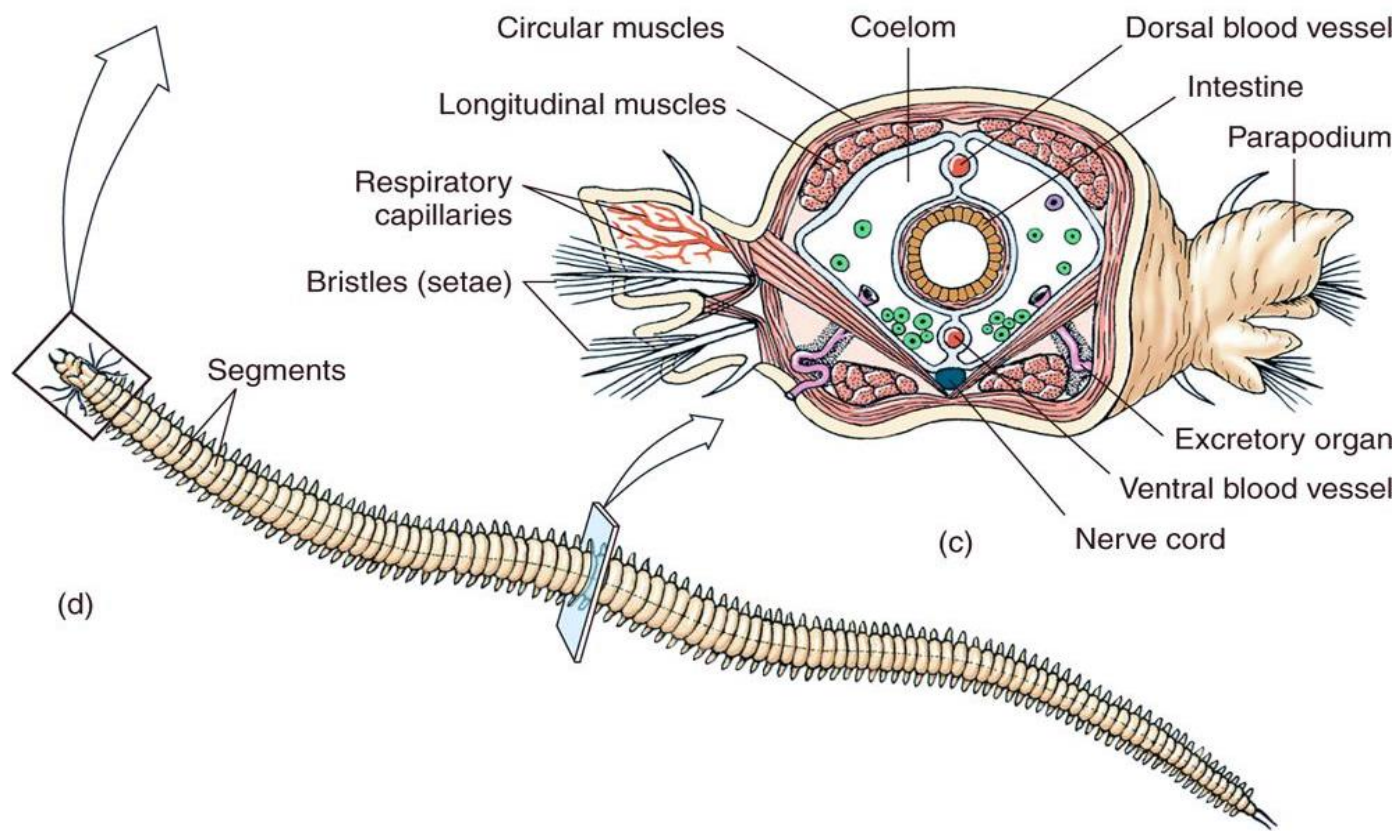
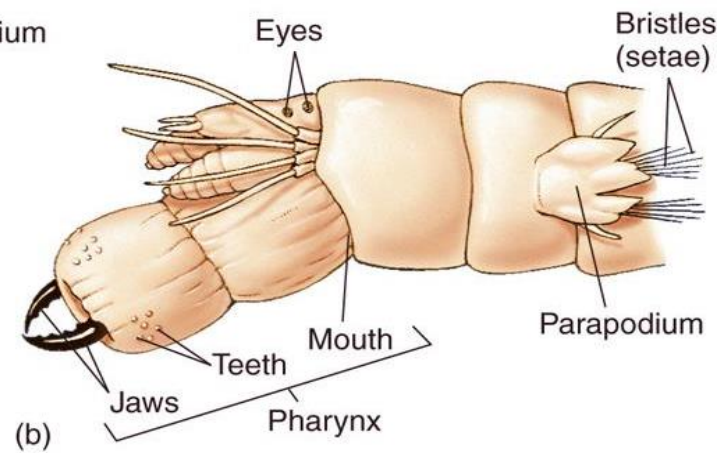
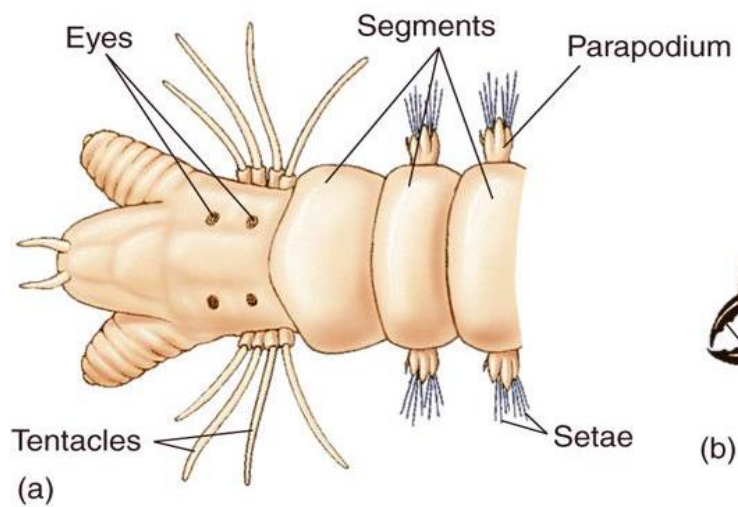
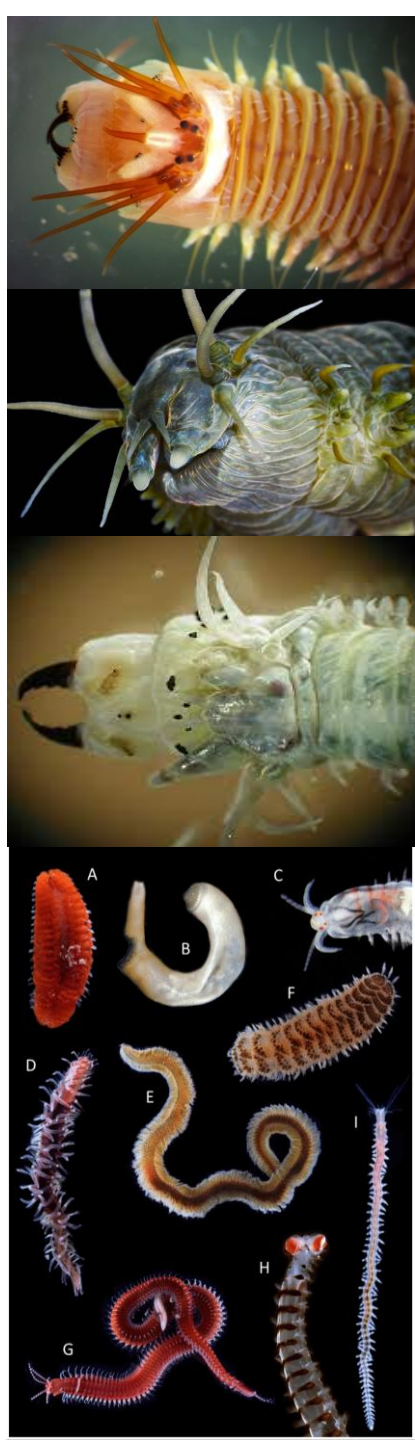


- **Features of Annelid worms**

- 1) **Complete digestive system**, from an anterior mouth to a posterior anus
- 2) **Coelom (体腔)**, a body cavity between the digestive tube and the external body wall
- 3) **Circulatory system** consisting of a series of tubes (vessels) filled with fluid (blood) to transport dissolved nutrients, oxygen, and waste products rapidly and efficiently
- 4) **CNS**
- 5) **Excretory system**: Consists of a pair of small tubes (nephridia, 后肾) in each segment

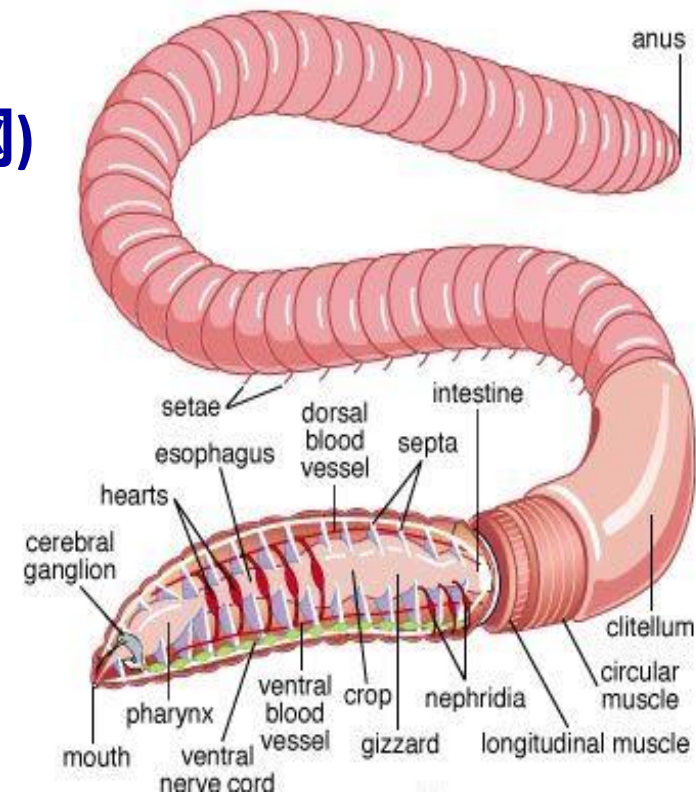
## **Example 1: Polychaetes** (Poly=many; chaeta=bristle, 多毛纲)

- **Largest group of annelids**, also most diverse, about 10,000 species
- Nearly exclusively marine, some in freshwater and brackish water
- Polychaeta worms are so named because most of their segments have **muscular flaps** called **parapodia (疣足)** (para=near; podia=feet)
- **Setae**: stiff bristles on parapodia, dig them into sand for locomotion
- Distinct **head**, some have a **proboscis** that can extend from their mouths to catch prey, feeding organ armed with small teeth or jaws on the tip
- Many have a planktonic larval stage called a **trochophore (担轮幼虫)**; this type of larvae is also seen in other invertebrate groups
- Can make a living in many habitats: mud, sand, sponges, live corals, and algae



## Example 2: Class Oligochaeta (寡毛纲)

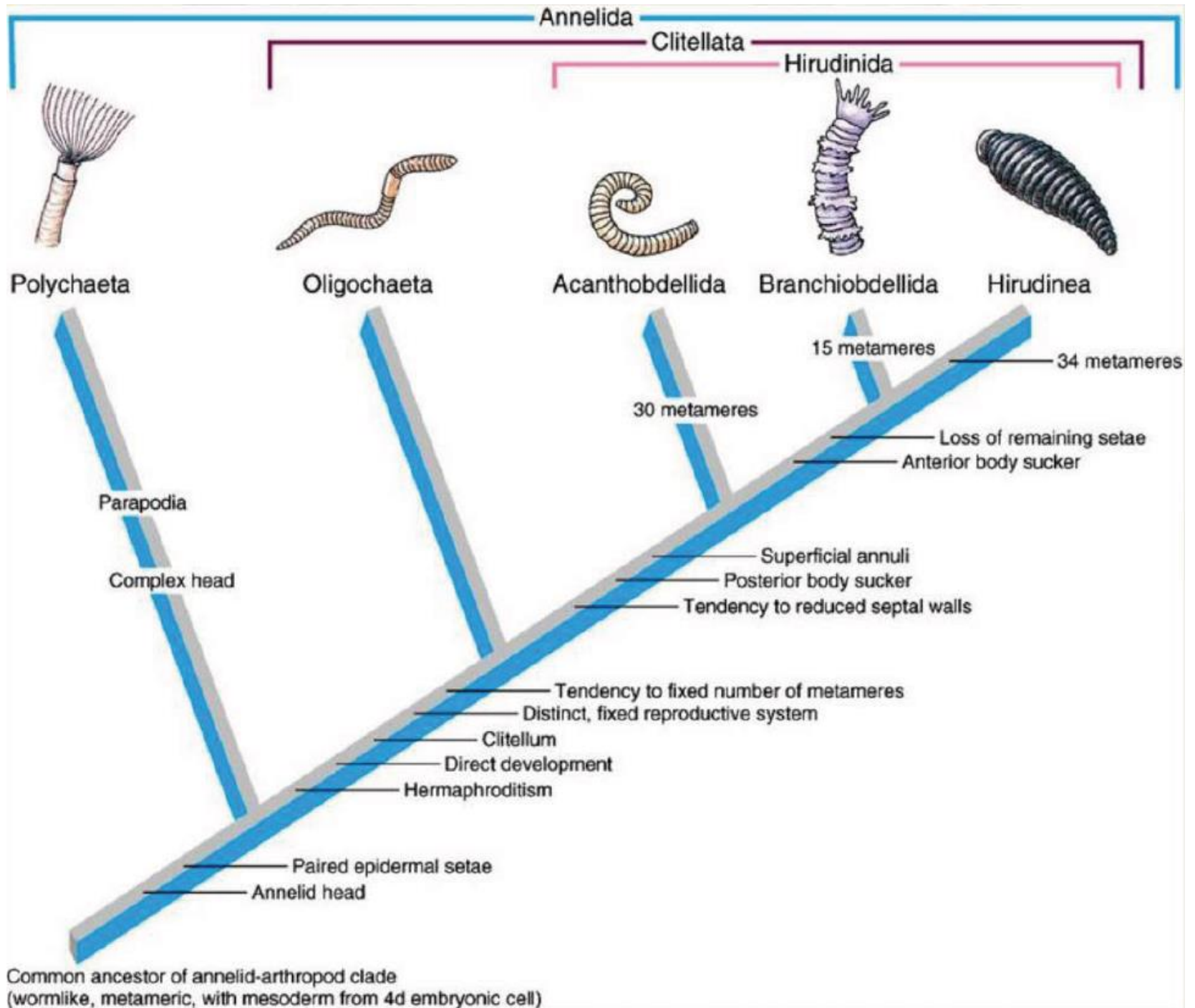
- Live in soil or fresh water
- Have **no parapodia**
- Have few setae
- Most familiar member is earth worm
- Closed circulatory system
- No specialized respiratory organs: oxygen and CO<sub>2</sub> diffuse directly through skin, gas exchange only happen only when skin is moist
- Eliminate cellular waste and excess water: Excretory tubules called nephridia, each segment (except first 3 and last 1) contain a pair



## Example 3: Leeches (蛭纲, Class Hirudinea)

- Most found in freshwater, some tropical species in moist terrestrial environment, few marine
- Marine species are parasitic and may prey on other invertebrates or fish
- Anterior and posterior suckers to hold prey
- Dorsoventrally flattened
- **Segmentation not as distinct as in other annelids**
- **No setae like in other annelids**





## • Major Phyla of Marine Invertebrates

- |                       |                            |
|-----------------------|----------------------------|
| 1. Sponges (海绵动物)     | 10. Molluscs (软体动物)        |
| 2. Cnidarians (腔肠动物)  | 11. Bryozoans (苔藓虫)        |
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| 9. Arthropods (节肢动物)  | 18. Lancelets (文昌鱼)        |

# 10: The Arthropods (节肢动物)

## 10.1 General Characteristics of Phylum:

- About 1 million species known, mostly marine
- About 75% of all animals on earth are arthropods
- Most marine species in a group of arthropods are called crustaceans (甲壳类)

Phylum

# Arthropods

Subphylum

**Prot**arthropoda  
(原节肢动物)

**Eu**arthropoda  
(真节肢动物)

Class

Onychophora  
(有爪纲)



- Arachnoida (蛛形纲)
- Insecta (昆虫纲)
- Myriapoda (多足纲)
- Merostomata (肢口纲)
- Crustacea (甲壳纲)



Copepoda  
(桡足亚纲)

Cirripedia  
(蔓足亚纲)

Malacostraca  
(软甲亚纲)

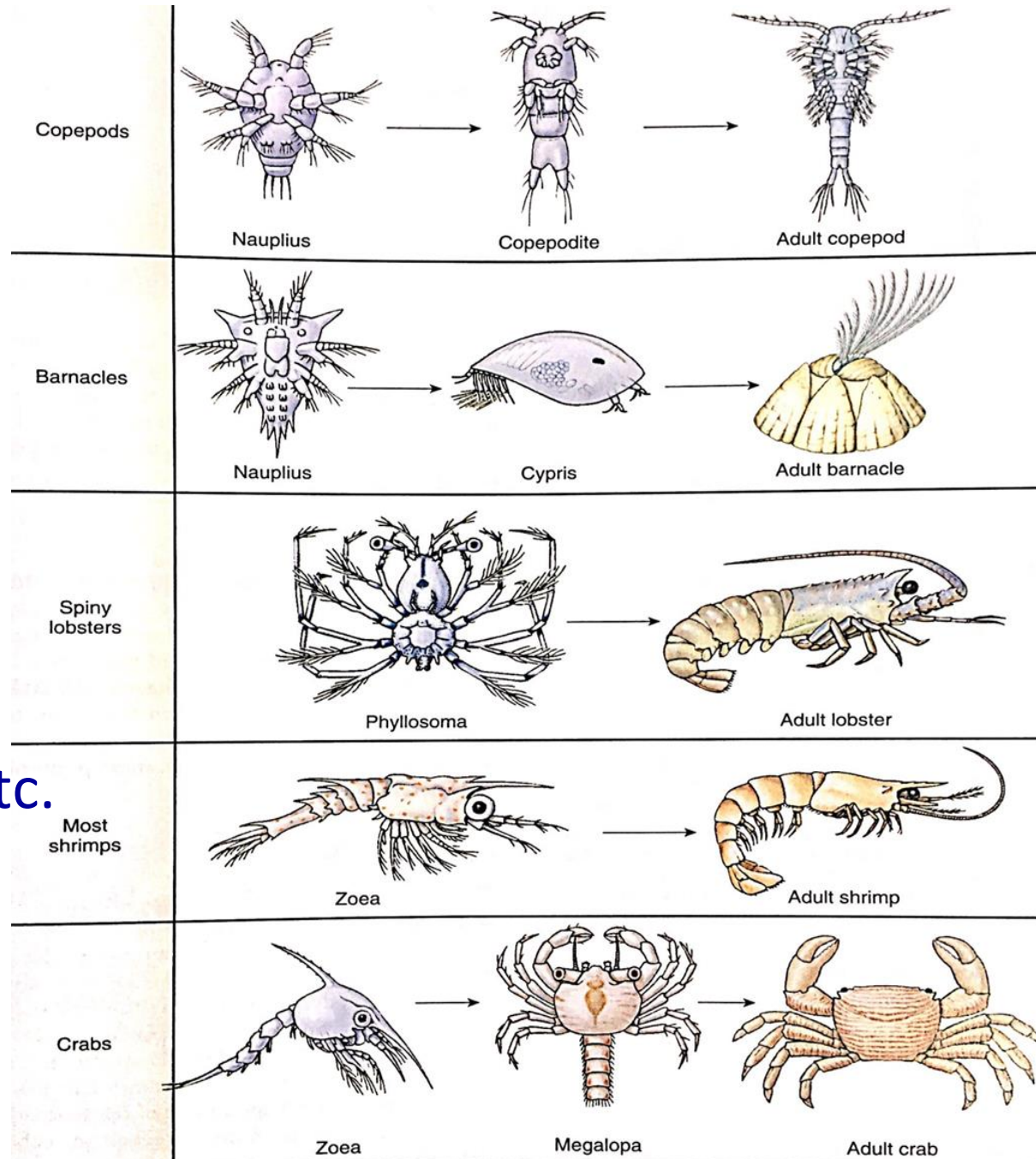


- Chitin (**几丁质**) **exoskeleton**: hard, but light and moderately flexible
- Since the skeleton is external, an arthropod must shed the shell to be able to grow. This process is called **molting (蜕皮)**, a soft new shell form underneath
- **Jointed appendages (具有关节的附肢)**
- Many divided into sections called **head, thorax (胸腔) and abdomen (腹腔)**
- **Specialized segmentation**: segments combined for specific functions
- **Specialized eye & sensory organs**: wide angle of vision

- **Specialized respiratory structures:** gills, used for gas exchange
- Can be filter feeders, scavengers (食腐) or carnivores
- More complex reproduction:
  - Many arthropods have complex behaviors including mating rituals (仪式)
  - Male transfers sperm directly to female to ensure reproductive success
  - Females can store sperm for fertilization at a later time
  - In some species, female will house eggs for a time until they are further developed

## 10.2 Marine Crustaceans

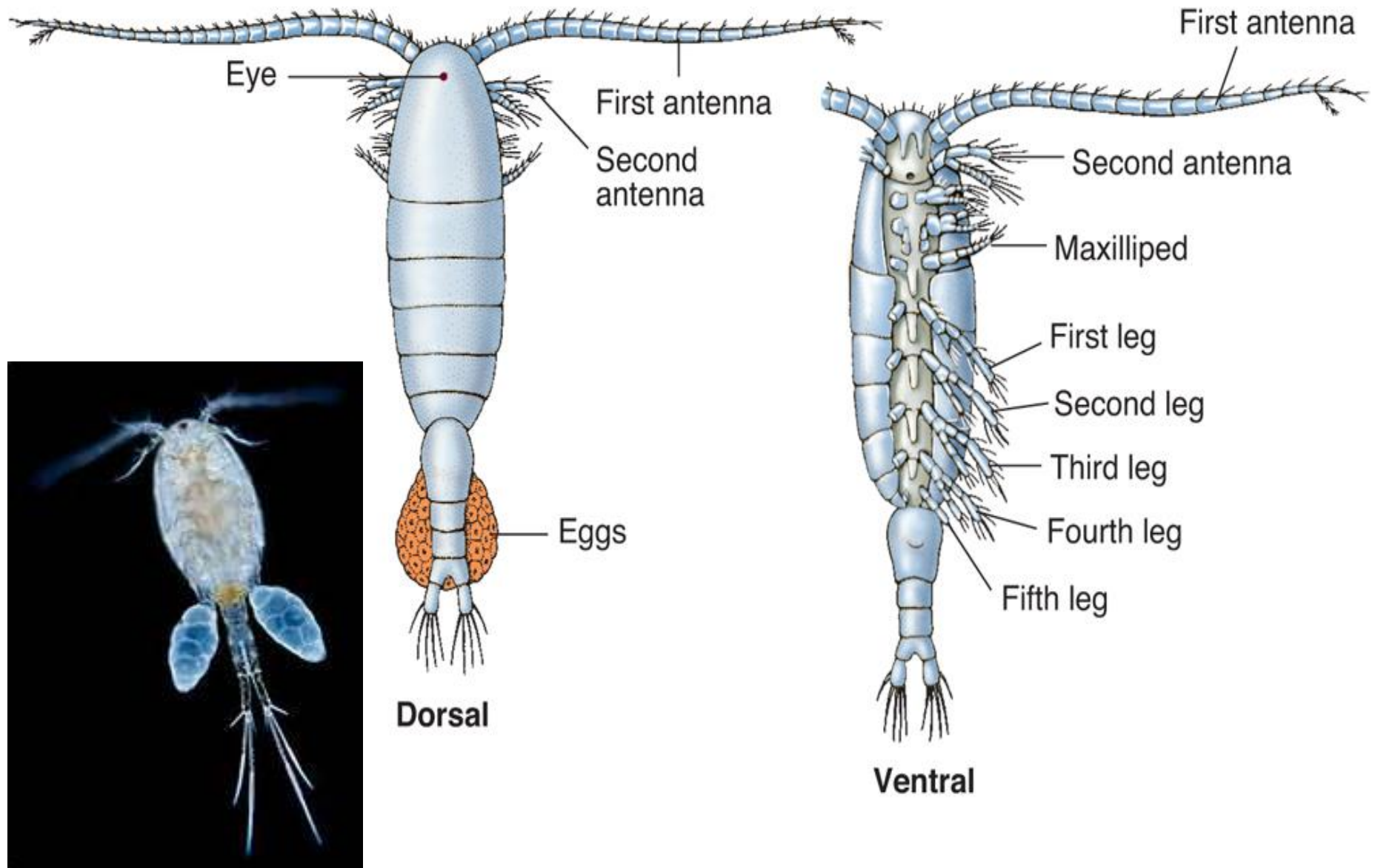
- 68,000 species
- copepods, barnacles, amphipods, isopods, shrimp, lobsters, crabs, etc.

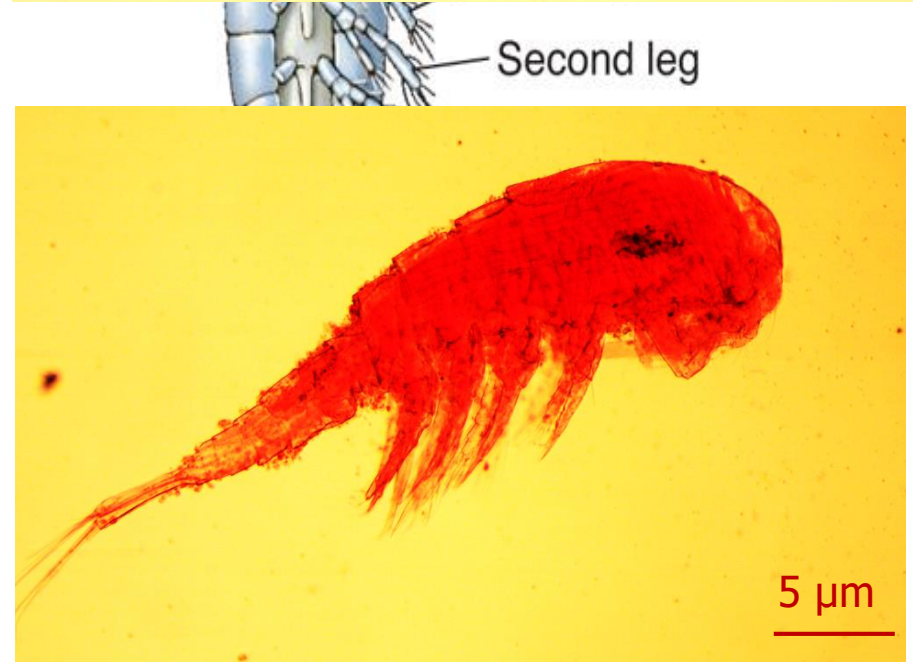
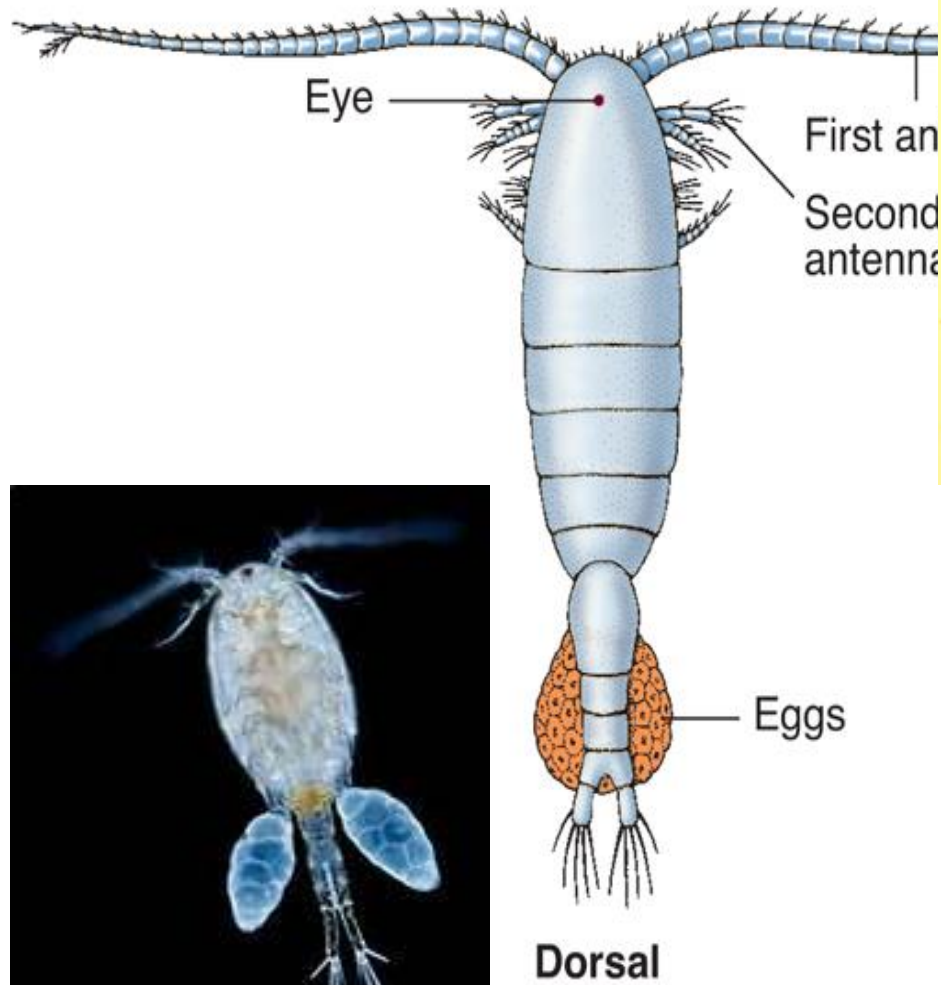


# 1) Copepods

- ❑ Extremely abundant and important in the plankton
- ❑ Use mouthparts to filter out or capture food
- ❑ Many planktonic species keep from sinking by using enlarged first pair of antennae to swim
- ❑ Many species are parasitic, some being so simplified that they look like little more than small bags of tissue

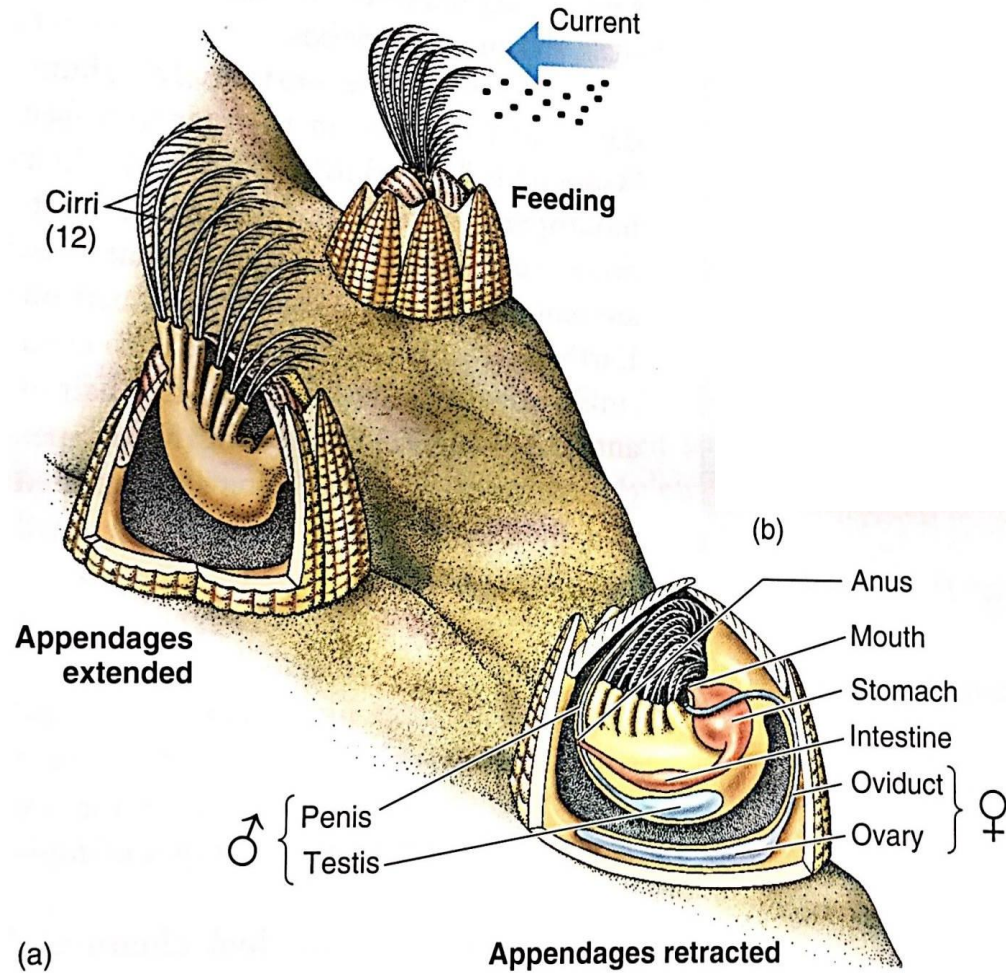
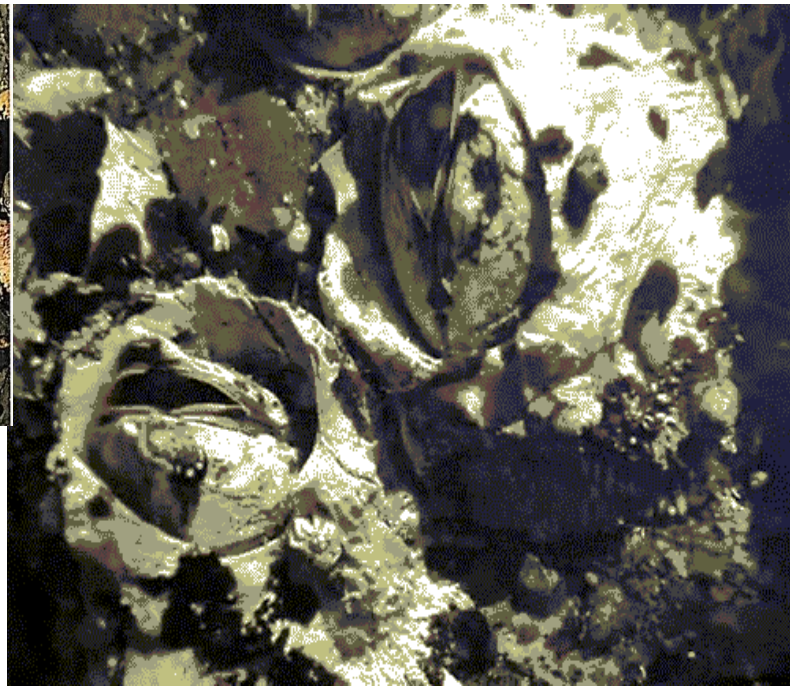






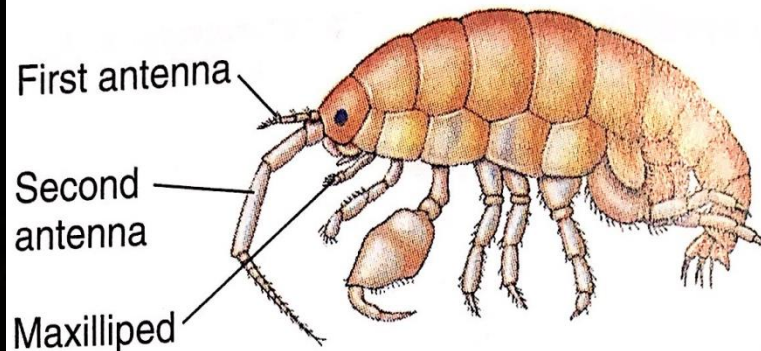
## 2) Barnacles (藤壶, 蔓足亚纲)

- ❑ Filter feeder usually live attached to surface, even living surfaces like whales, crab, bivalve (双壳类)
- ❑ Some are among the most important types of **fouling organisms** (腐蚀生物)
- ❑ Body is enclosed by **heavy and calcareous plates**
- ❑ The plates on the upper surface open to allow the **feathery filtering appendages (cirri)**, which are actually the legs, to sweep the water
- ❑ **All barnacle have typical crustacean larvae**, which swim and attach to surfaces before metamorphosing into adults



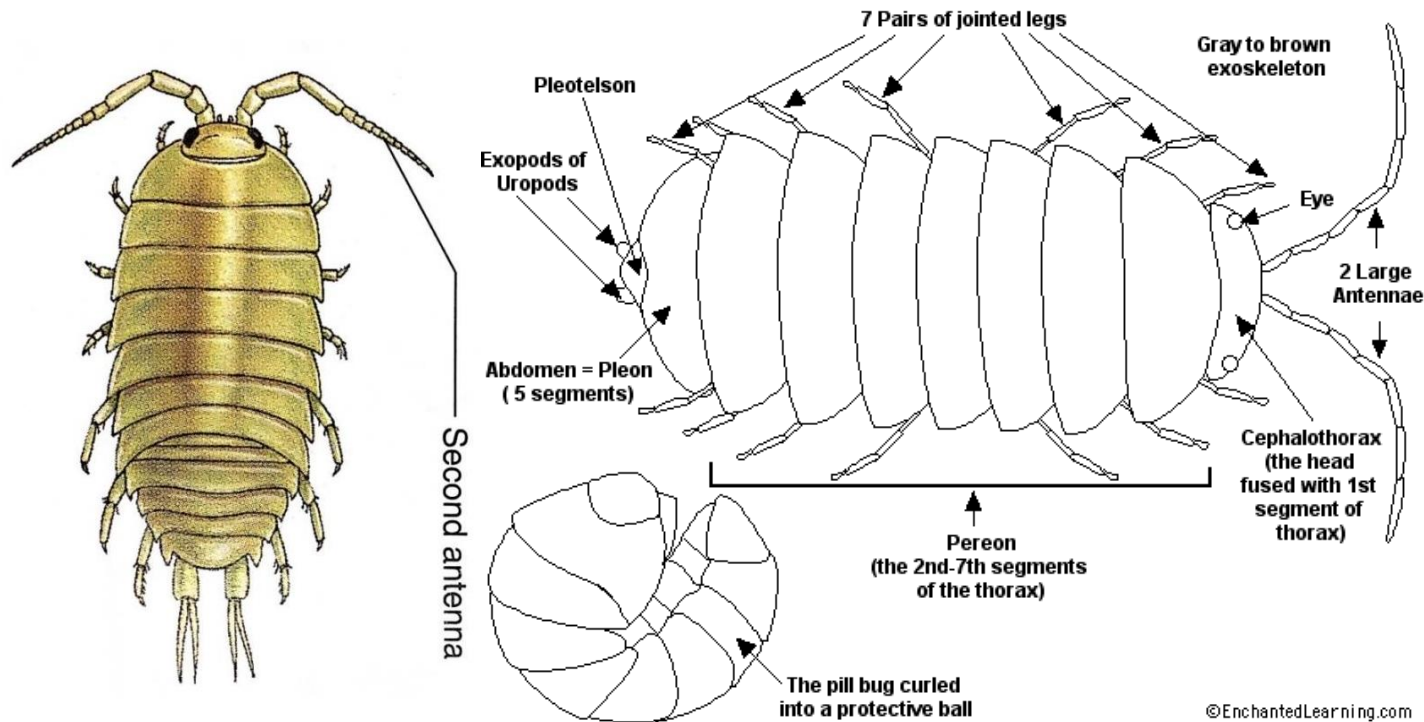
### 3) Amphipods (片脚类)

- ❑ Over 5,000 species, mostly marine
- ❑ Small crustaceans with a curved body that is **flattened sideways**
- ❑ Most under 2 cm in length, deep-water species are larger
- ❑ The head and tail typically curve downward, and the appendages are specialized according to function
- ❑ Beach hoppers/strong jumpers, can crawl among seaweeds or burrow in the skin of whales



## 4) Isopods (等足类)

- ❑ Similar size and living environment like Amphipods
- ❑ Main part of the body has legs that are similar to each other and the body is dorsoventrally flattened and thus has a flat back
- ❑ Some species can be parasites of fishes and other, e.g. fish lice

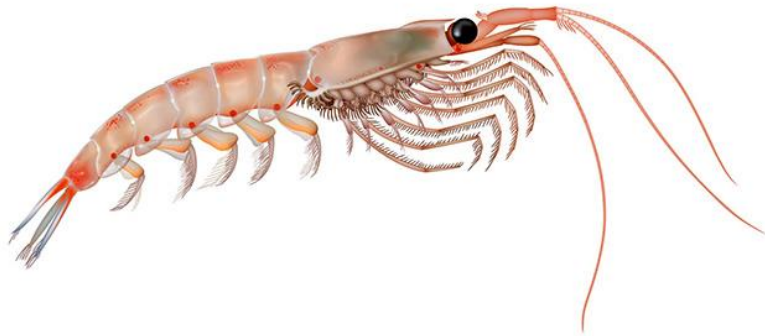


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## 5) Krill (磷虾) or euphausiids

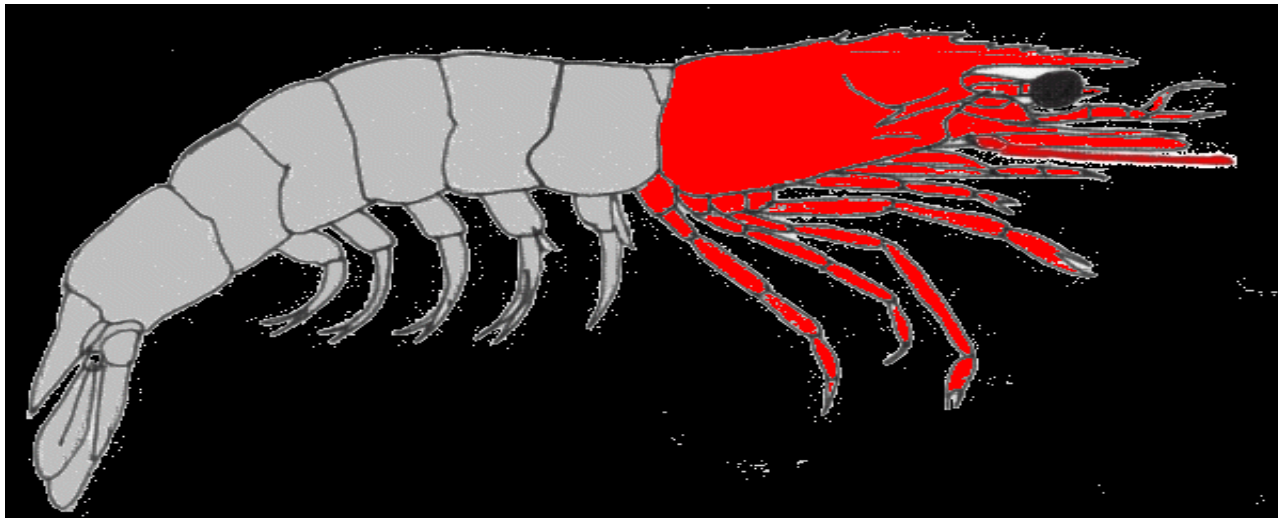
- ❑ Planktonic, shrimp-like crustaceans up to 6 cm
- ❑ The head fused with body segments to form a distinctive carapace, covers the anterior half of the body like armor
- ❑ Most are filter feeders feeding on diatoms and other plankton
- ❑ Extremely common in polar waters, aggregating in gigantic group of billions of individuals, others are living in deep sea
- ❑ Exclusive food for many Antarctic whales, penguins, and fishes



## 6) **Decapods (十足目)**: shrimp, lobster and crab

- Around 10,000 species, the **largest group of crustaceans**, also the **largest crustaceans in size**
- **Five pairs** of legs, or **pereopods/periopods (步足)**, the first of which is heavier and usually has claws used for feeding and defense; appendages on abdomen are called **pleopods (腹肢)**
- **Paired maxillipeds (颚/颚片)**, which are closer to the mouth, turned forward, and specialized to sort out food and push it toward the mouth, including **maxilla (小颚)** and **mandible (大颚)**

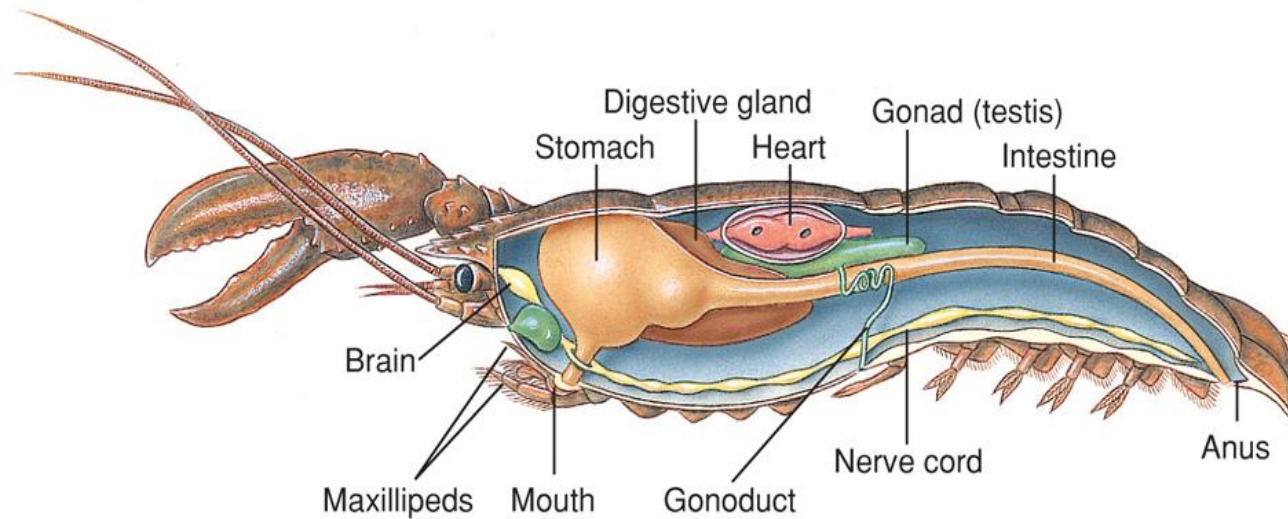
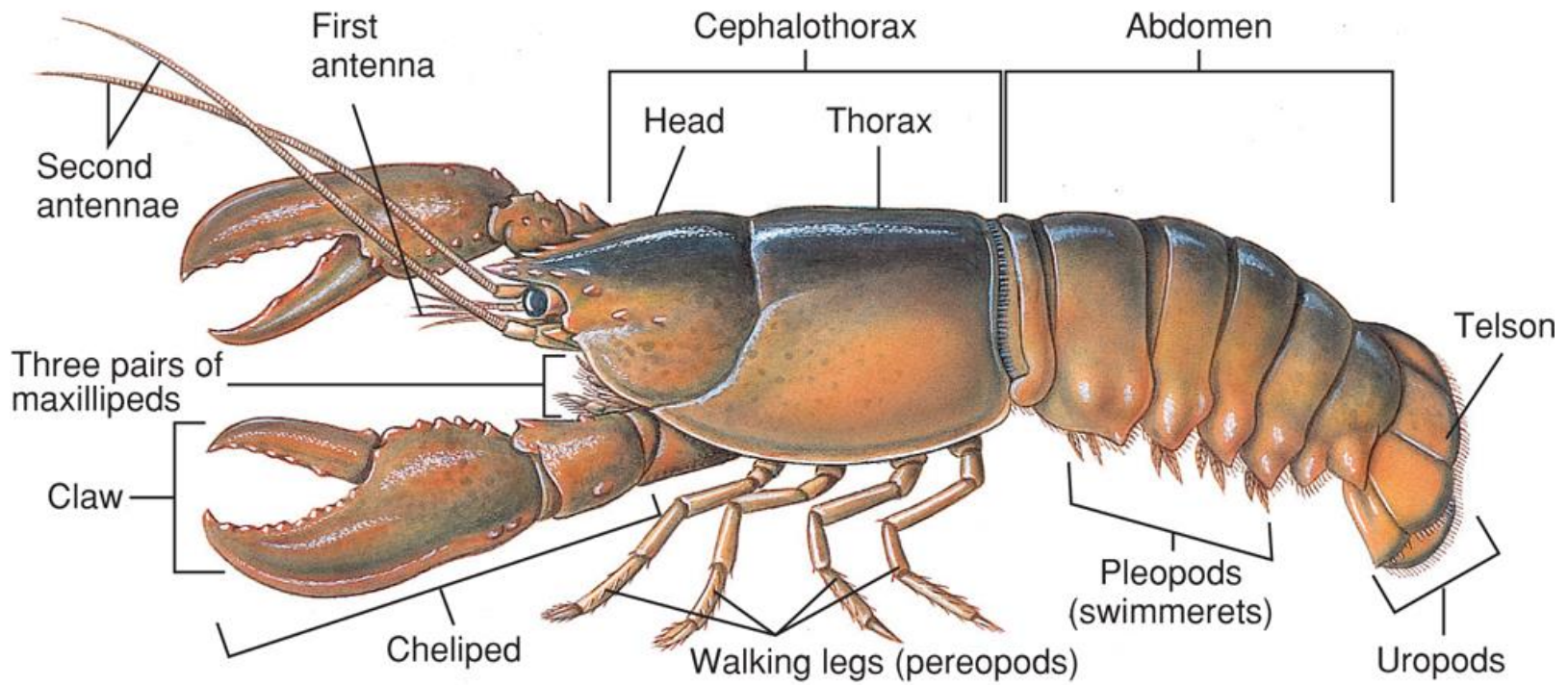
- ❑ 2 pairs antennae (触须)
- ❑ Head and thorax fused into a single unit called a cephalothorax (头胸部) ; entire external body is called the carapace (甲壳)
- ❑ Large array of appendages specialized for different functions: pinchers (钳子) /chelate (螯) on crabs  
swimmerettes (游泳器) on the underside of shrimp  
hold developing eggs, etc.



## Example 1: Shrimps

- Have laterally compressed bodies with distinct and elongated abdomens (we call it 'tail' and like to eat)
- Shrimp are typically scavengers (食腐), specialists in the feeding on bits of detritus on the bottom
- Many colorful shrimps live on the surface of other invertebrates or remove parasites from the skin of fish
- Ghost and mud shrimps burrow in muddy bottoms





## Example 2: Lobsters

Lobsters are mostly nocturnal (夜间活动的) and hide during the day in rock or coral crevices (裂缝), scavengers and predators that crush mollusks and sea urchins

